

MOHAMED RIZK

Computer Science & AI Student | Backend Developer | Data Science & Computer Vision
Cairo, Egypt | +20 102 324 2250 | mohamedrezk66091@gmail.com | [Linkedin](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Senior-year Computer Science & AI student at the British University in Egypt (GPA: 3.4), combining strong backend development skills with hands-on experience in data science and computer vision. Built and deployed full-stack Java desktop applications backed by MySQL, trained object-detection models (YOLOv5/YOLOv8) for real-world safety applications, and conducted a two-phase epidemiological data analysis pipeline on CDC COVID-19 surveillance data. Actively seeking internships or entry-level roles in software engineering, data science, or AI.

EDUCATION

British University in Egypt (BUE)

BSc (Hons) Informatics & Computer Science — Artificial Intelligence | GPA: 3.4

El-Shorouk, Egypt

Expected: 2027

Relevant Coursework: Visual Computing & Image Processing · Introduction to Data Science · Machine Learning · Software Engineering · Data Structures & Algorithms · Databases · Operations Research · Logic & Artificial Intelligence · Cybersecurity · Distributed Systems

TECHNICAL SKILLS

Programming	Python, Java, C/C++, SQL
AI / ML	YOLOv5, YOLOv8, Ultralytics, PyTorch, Scikit-learn, Transfer Learning
Computer Vision	OpenCV, Image Preprocessing, Data Augmentation, Object Detection
Data Science	Pandas, NumPy, Matplotlib, Seaborn, Hypothesis Testing, Web Scraping (BeautifulSoup, requests)
Backend Dev	OOP, JDBC, REST API Simulation, Software Architecture, MySQL, MS Access
Frontend Dev	HTML, CSS, JavaScript, Bootstrap, AOS.js, Streamlit
Tools & Platforms	Git, GitHub, Jupyter Notebook, VS Code, NetBeans, Figma, Excel, Swing GUI, Kaggle, Roboflow
Methodologies	Agile Basics, SDLC, CRISP-DM Data Science Workflow

PROJECTS

Hard Hat & Safety Gear Detection (Computer Vision) [Python](#) · [OpenCV](#) · [YOLOv5](#) · [YOLOv8](#) · [Ultralytics](#) · [PyTorch](#) · [Streamlit](#) · [Roboflow](#) 2025 – 2026

Module: Visual Computing & Image Processing (25CSAI02I) — 40% of course marks | BUE, Semester 2

- Led Subteam B independently — sourced and preprocessed a 9-class PPE Safety dataset (~1,420 images) from Roboflow Universe covering hardhats, vests, gloves, shoes, and their absence.
- Built a full image preprocessing pipeline: letterbox resizing to 640×640, BGR→RGB conversion, pixel normalisation, and Gaussian blur noise reduction.
- Implemented 7 data augmentation techniques (horizontal flip, rotation $\pm 15^\circ$, translation, random crop, brightness jitter, HSV jitter, Mosaic) to improve model generalisation.
- Trained YOLOv5s and YOLOv8s with COCO pretrained weights (50 epochs, batch 16); expected mAP@0.5 ~0.78–0.84; compared anchor-based vs anchor-free detection paradigms.
- Co-developed a Streamlit web app enabling interactive model comparison across 4 trained models with confidence threshold slider, bounding box overlay, and metrics dashboard.

COVID-19 Case Surveillance — Two-Phase Data Science Analysis [Python](#) · [Pandas](#) · [NumPy](#) · [Matplotlib](#) · [Seaborn](#) · [BeautifulSoup](#) · [SciPy](#) 2025 – 2026

Module: Introduction to Data Science (25CSAI01I) | BUE, Semester 2

- Phase 1: Analysed 2,500 individual CDC COVID-19 records across 12 research questions covering age-mortality trends, sex-based outcome differences, ICU severity, and monthly epidemic waves.
- Phase 2: Designed and implemented a BeautifulSoup web scraper to collect weekly state-level case/death data from the CDC open data portal; performed full data cleaning, negative-death correction, state-name mapping, and inner-join integration with Phase 1 data across 20 matched US states.
- Answered 3 assigned Phase 2 questions using a dual-axis state comparison chart, multi-series time-series line chart, and annotated correlation heatmap (seaborn) across all integrated metrics.
- Conducted Welch's t-test with Cohen's d effect size to test age-stratified vaccine-uptake hypothesis ($p = 0.391$; failed to reject H_0 ; negligible effect $d = -0.038$).
- Applied full CRISP-DM workflow: data collection → cleaning → integration → EDA → visualisation → statistical inference, demonstrating end-to-end data science competency.

Pharmacy Management System [Java](#) · [MySQL](#) · [Swing GUI](#)

- Developed a desktop CRUD application with search, update, delete, and auto-fill; dark/light theme toggle integrated into personal portfolio.

Barber Shop Appointment System [Java](#) · [MySQL](#) · [JDBC](#)

- Built a scheduling system with calendar input and real-time JDBC data handling for persistent storage.

Student Record Management System [Java](#) · [MySQL](#)

- Designed a modular student information system using OOP principles with input validation and clean architecture.

RESTful API Simulation Tool [Java \(Console\)](#)

- Simulated HTTP methods (GET, POST, PUT, DELETE) to model backend API behaviour and REST stateless communication.

Gym Website [HTML](#) · [CSS](#) · [JavaScript](#) · [AOS.js](#)

- Designed and developed a fully responsive animated website with scroll-based AOS.js animations.

School Management System & Attendance / Prison Systems [Java](#) · [Swing](#) · [File I/O](#) · [MS Access](#)

- Built desktop record-management applications with file-based and MS Access data persistence, demonstrating structured database schema design.

EXPERIENCE

Activation Team Member — EL-Araby Group

- Supported marketing activation campaigns; developed strong public engagement and communication skills.

Event Usher — Tazkarti · Red Bull · Lipton · TTL · P+US

- Managed crowd coordination and ensured smooth, high-quality attendee experiences at large-scale events.

TRAINING & CERTIFICATIONS

- **Sprints × Microsoft Summer Camp — Web Development** | 40 hrs | ID: SPR-R77X4V
- **EgyptAir Summer Training** | July 2025
- **ITI Training Program** | August 2025
- **Google 5-Day Intensive Course** (In Progress)
- **Backend Development Course** (In Progress)
- **SEO Analysis** — Career 180 Platform, January 2023
- **Stock Market & Investment Workshop** — Ostoul Capital × BUE, 2025
- Self-study: Full-Stack Software Development — Python, JavaScript, HTML/CSS, Finance

ACTIVITIES & LEADERSHIP

- Organiser, First-Year Experience Committee — British University in Egypt (BUE)
- PR Coordinator, Student Union (SU) — British University in Egypt (BUE)

LANGUAGES

Arabic: Native **English:** Fluent **German:** Intermediate